Computer Science - Undergraduate Level Assessment 14 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Which of the following sorting algorithms has average-case and worst-case running times of O(n log n)?
   (a) Bubble sort
   (b) Insertion sort
   (c) Merge sort
   (d) Quicksort

2. Of the following, which gives the best upper bound for the value of f(N) where f is a solution to the recurrence f(2 N + 1) = f(2 N) = f(N) + log N for N >= 1, with f(1) = 0?
   (a) O(log N)
   (b) O(N log N)
   (c) O(log N) + 0(1)
   (d) O((log $N)^2$)

3. Let x = 8. What is x>>1 in Python 3?
   (a) 3
   (b) 4
   (c) 2
   (d) 8

4. In the Internet Protocol (IP) suite of protocols, which of the following best describes the purpose of the Address Resolution Protocol?
   (a) To translate Web addresses to host names
   (b) To determine the IP address of a given host name
   (c) To determine the hardware address of a given host name
   (d) To determine the hardware address of a given IP address

5. Of the following page-replacement policies, which is guaranteed to incur the minimum number of page faults?
   (a) Replace the page whose next reference will be the longest time in the future.
   (b) Replace the page whose next reference will be the shortest time in the future.
   (c) Replace the page whose most recent reference was the shortest time in the past.
   (d) Replace the page whose most recent reference was the longest time in the past.

True / False (10 marks)

Answer True or False

6. True or False: O(1) represents constant time complexity, making it the smallest asymptotic notation compared to O(n), O(n^2), and O(log n).

7. True or False: If no errors are found during testing, it ensures that all preconditions are correct.

8. True or False: The condition a $<$ c is sufficient to ensure that the expression a $<$ c || a $<$ b && !(a == c) evaluates to true.

9. True or False: In Python 3, converting a list to a set removes duplicates and returns unique elements in an unordered collection.

10. True or False: The decimal number 0.5 has an exact representation in binary notation as 0.1.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to find the number which occurs for odd number of times in the given array.

12. Write a function to sort each sublist of strings in a given list of lists using lambda function.

End of Paper